

Midpoint preservation on a symmetric lens

MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens

theorem

Identifies the centers of two lenses without prescribing images of their foci.

Statement

Let E, F be real normed spaces, $x, y \in E$, $x', y' \in F$, and $r \in \mathbb{R}$. Put $\Lambda_r(x, y) = \{z : \|z - x\| \leq r, \|z - y\| \leq r\}$, and define the target lens in F similarly. Suppose $\|x - y\|/2 \leq r$ and $\|x' - y'\|/2 \leq r$. Every bijective isometry $f : \Lambda_r(x, y) \rightarrow \Lambda_r(x', y')$ satisfies

$$f\left(\frac{x + y}{2}\right) = \frac{x' + y'}{2}.$$

Assumptions

The two real normed spaces, both half-distance inequalities, and the bijective isometry between the lenses are given. The radius is common to both lenses. The foci need not themselves belong to the lenses, and no equation for their images is assumed.

Conclusion

The midpoint of the source foci is sent to the midpoint of the target foci.

Proof route

Verify membership, boundedness and reflection invariance for these exact sets, and then apply center transport.

▼ Proof steps

1. For $m = (x + y)/2$, one has $\|m - x\| = \|m - y\| = \|x - y\|/2 \leq r$. Hence $m \in \Lambda_r(x, y)$ by Midpoint membership. The same calculation places $m' = (x' + y')/2$ in the target lens.
2. The source lens is contained in $\overline{B}(x, r)$ and is therefore bounded. For z in that lens, reflection about m is $r_m(z) = x + y - z$, and

$$\|r_m(z) - x\| = \|z - y\| \leq r, \quad \|r_m(z) - y\| = \|z - x\| \leq r.$$

This verifies Reflection invariance of a lens for the source; the identical identities with primes verify the target condition.

3. Apply Center transport to $s = \Lambda_r(x, y)$, $t = \Lambda_r(x', y')$, centers m, m' and this same f . Step 1 supplies center membership and Step 2 supplies boundedness and both invariances. Its output $f(m) = m'$ is exactly the asserted equation.

Main citations

- Exact theorem and proof
- Lens definition
- Midpoint membership
- Reflection invariance of a lens
- Center transport

Lean source signature (exact)

```
theorem map_midpoint_of_symmetricLens
  {x y : E} {x' y' : F} {r : ℝ}
  (hxy : 2-1 * dist x y ≤ r) (hxy' : 2-1 * dist x' y' ≤ r)
  (f : symmetricLens x y r ≈i symmetricLens x' y' r) :
  ((f (midpoint ℝ x y, midpoint_mem_symmetricLens hxy) :
    symmetricLens x' y' r) : F) = midpoint ℝ x' y'
```

In the source	Mathematical meaning
$\{x\ y : E\} \{x'\ y' : F\} \{r : \mathbb{R}\}$	The two source foci x, y , two target foci x', y' , and common real radius r ; E, F are real normed spaces.
$(hxy : 2^{-1} * \text{dist } x\ y \leq r)$	The bound $\ x - y\ /2 \leq r$ which puts $(x + y)/2$ in the source lens.
$(hxy' : 2^{-1} * \text{dist } x'\ y' \leq r)$	The corresponding target bound $\ x' - y'\ /2 \leq r$.
$\text{symmetricLens } x\ y\ r \approx_i \text{symmetricLens } x'\ y'\ r$	A bijective isometry of the two intersections of closed balls. This does not require $f(x) = x'$ or $f(y) = y'$.
$\text{midpoint_mem_symmetricLens } hxy$	The membership proof obtained from the source half-distance bound, allowing f to be evaluated at $(x + y)/2$.
$((f (\text{midpoint } \mathbb{R}\ x\ y, \text{midpoint_mem_symmetricLens } hxy) : \text{symmetricLens } x'\ y'\ r) : F)$	The same image point $f((x + y)/2)$, viewed in the ambient target space F .
$= \text{midpoint } \mathbb{R}\ x'\ y'$	The conclusion that this image equals $(x' + y')/2$.

Exact source and proof.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: `MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens`

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